Table 1: Requirements for carriage of liquefied hydrogen in bulk

a	b	c	d	e	f	g	h	i
Product name	Ship type	Independent tank type C required	Control of vapour space within cargo tanks	Vapour detection	Gauging	Special requirements		
Hydrogen	2G	-	-	F	C	See table 2 and, either corresponding table 4 or table 5, as appropriate for the type of cargo containment systems		

SOURCE: MSC.566(109).pdf

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Hydrogen 2G - - F C See table 2 and, either corresponding table 4 or table 5, as appropriate for the type of cargo containment systems

Table 2: Special requirements for carriage of liquefied hydrogen in bulk

No.	Special requirement	Related hazard
A-1	Requirements for materials whose design temperature is lower than -165°C should be agreed with the Administration, paying attention to appropriate standards. Where minimum design temperature is lower than -196°C, property testing for insulation materials should be carried out with the appropriate medium, over a range of temperatures expected in service.	Low temperature (see 4.2.1)
A-2	Materials of construction and ancillary equipment such as	

insulation should be resistant to the effects of high oxygen concentrations caused by condensation and enrichment at the low temperatures attained in parts of the cargo system (refer to the requirement for nitrogen). This special requirement is applied
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low temperatures attained in parts of the cargo system (refer to the requirement for nitrogen). This special requirement is applied
to all locations where contact with condensed oxygen is anticipated under normal conditions and foreseeable single failure scenarios.
Low temperature
(see 4.2.2)
A-3 For cargo pipes containing liquid hydrogen and cold hydrogen vapour, measures should be taken to prevent the exposed surfaces from reaching -183°C. For places where preventive measures against low temperature are not sufficiently effective, such as cargo m anifolds, other appropriate measures such as ventilation which avoids the formation of highly enriched oxygen and the installation of trays recovering liquid air may be permitted in lieu of the preventive measures. Insulation on liquid hydrogen piping syst ems exposing to air should be of non -combustible material and should be designed to have a seal in the outer covering to prevent the condensation of air and subsequent
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material and should be designed to have a seal in the outer covering to prevent the condensation of air and subsequent
oxygen enrichment within the insulation.
Low temperature
(see 4.2.2)
A-4 Appropriate means, e.g. filtering, should be provided in cargo piping systems to remove impure substances condensed at low temperature.
Low temperature
(see 4.2.3)
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No. Special requirement Related hazard
A-5 Pressure relief systems should be suitably designed and constructed to prevent blockage due to formation of water or ice.
Low temperature
(see 4.2.4)
A-6 At places where contact with hydrogen is anticipated, suitable

materials should be used to prevent any structural deterioration owing to hydrogen embrittlement and degradation of strength and
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materials should be used to prevent any structural deterioration owing to hydrogen embrittlement and degradation of strength and
fatigue properties due to continual exposure to hydrogen , as necessary.
Hydrogen embrittlement (see 4.3)
A-7 Double tube structures ensuring no leakage, or fixed hydrogen detectors being capable of detecting a hydrogen leak, should be provided for confined places where leakage of hydrogen may occur, such as cargo valves, flanges and seals.
Susceptibility to leakage (see 4.4.2)
A-8 Helium or a mixture of 5% hydrogen and 95% nitrogen should be used as the tightness test medium for cargo tank and cargo piping.
Susceptibility to leakage (see 4.4.3)
A-9 The amount of carbon dioxide carried for a carbon dioxide fire-extinguishing system should be sufficient to provide a quantity of free gas equal to 75% or more of the gross volume of the cargo compressor and pump rooms in all cases.
Fire by Hydrogen (see 4.7.3)
Wide range of

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the cargo compressor and pump rooms in all cases.
Fire by Hydrogen (see 4.7.3)
Wide range of flammability limits (see 4.10)
A-10 When deterioration of insulation capability by single damage is possible, appropriate safety measures should be adopted taking into account the deterioration.
High pressure (see 4.8)
A-11 Appropriate measures should be provided to prevent vents becoming blocked by accumulations of ice formed from moisture in the air.
Low temperature

(see 4.2.2)
A-12 Due consideration should be given to means for handling boil-off gas.
High pressure
(see 4.8)
A-13 Due consideration should be given to static electricity associated with rotating or reciprocating machinery including the installation of conductive machinery belts and precautionary measures incorporated in operating and maintenance procedures , in addition to the bonding of tanks, piping and equipment required by paragraph 5.7.4 of the Code.

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addition to the bonding of tanks, piping and equipment required by paragraph 5.7.4 of the Code.
Anti-static clothing and footwear, and a portable hydrogen detector should be provided for each crew member working in the cargo area.
Static electricity
(see 4.9.2)
A-14 The cargo operation manuals required in paragraph 18.2 of the Code should include limitations of various operations in relation to environmental conditions.
Wide range of
flammability limits
(see 4.10)
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